ClimateKG — Surfacing Knowledge

A community knowledge graph:
Based on FAIR Data & UNESCO Open Science Principles

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About ClimateKG

- Climate Knowledge Graph (ClimateKG) is a yearlong R&D project to create a knowledge graph of the *IPCC Sixth Assement Report (AR6)*.
- The ClimateKG knowledge graph is built using Wikibase and MediaWiki.
- The goal is to be a resource for:
  - data science use, for
  - citizen science activities, and to
  - distribute climate science metadata to Wikidata.

The project has been funded by **TIB Innovation Fund** and is made in cooperation with #semanticClimate community (India, Germany, UK, +).



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What are the IPCC Reports?



> It is a survival guide for humanity. As it shows, the 1.5-degree limit is achievable.
> - UN Secretary-General António Guterres (2023)

- A multilateral panel of 195 nations that reviews the global scientific literature to map out climate scenarios for the next 100 years combining science, policy, and politics.
- AR6 (≈ 5-8 year cycle since 1988): 932 authors; 7 Reports; >8 million words; 10,047 pages; 48,400 citations; 66,834 data sets; 2,136 images; 1,910 Acronyms; 920 Glossary items; 5 languages+ (partial).*

* Oldenbourg, Laura, and Simon Worthington. “IPCC AR6 Quantification Summary”. Climate Knowledge Graph, November 4, 2025. 10.5281/zenodo.17521936.

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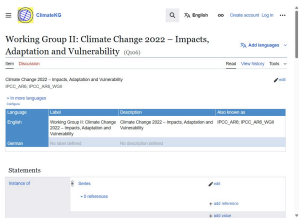
ClimateKG Project Deliverables

Transform PDF & web corpus, and web databases into a knowledge graph with a reusable pipeline made by ClimateKG:

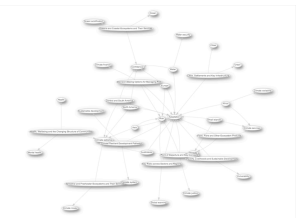
Full text & publishing



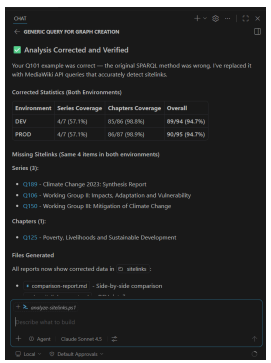
Metadata



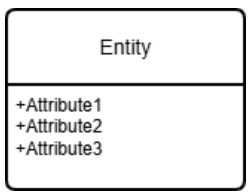
API / SPARQL Endpoint



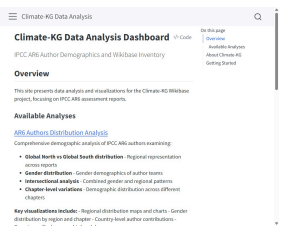
Claude & LLMs IDE integration



Schemas / Ontologies: DataCite, OpenAlex, Basic Formal Ontology (BFO), etc



Data workbench: JupyterLab / Quarto



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Problem: Public Trust in Climate Science Requires It's at Their Fingertips!

- **Headline issues:**
  - Search and SEO limited
  - Not easy to reuse
  - Formats of Web CMS HTML and PDF not suitable for data science
  - Referenced materials not interlinked — only manual tracking possible
  - Parts not easily available — Methodology, Supplementary material, etc
- **Examples:**
  - Links only to top levels: Reports, Glossary or Author websites, etc.
  - Glossary terms have no links to reports
  - Author lists have no link to chapters
  - Figures listed on web pages with no DOIs
  - Citations on websites and not research repositories

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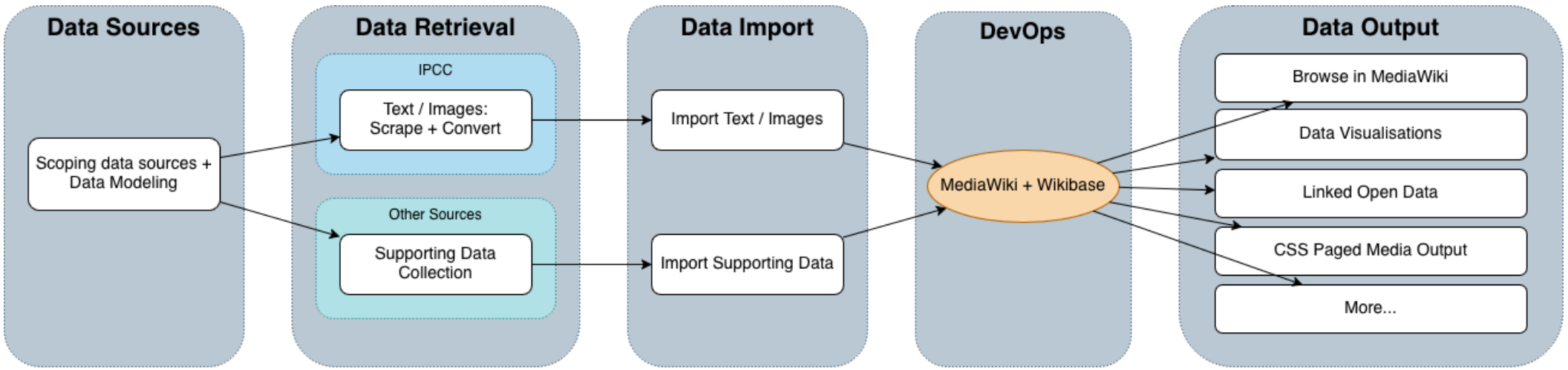
Solution: Apply the *vision* of the Semantic Web

> Link all of the parts and make machine readable
>
> — AKA a Knowledge Graph

- **Linked Data**: Interlinked data with unique identifiers and standard formats
- **FAIR Data**: Findable, Accessible, Interoperable, and Reusable

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Overview of work packages



- From **Siloed Data** to **FAIR Data**
- **Finding** the sources → **retrieving** the data → **storing** it → preparing different ways of **outputting** and **visualizing** it and **making data available for reuse**

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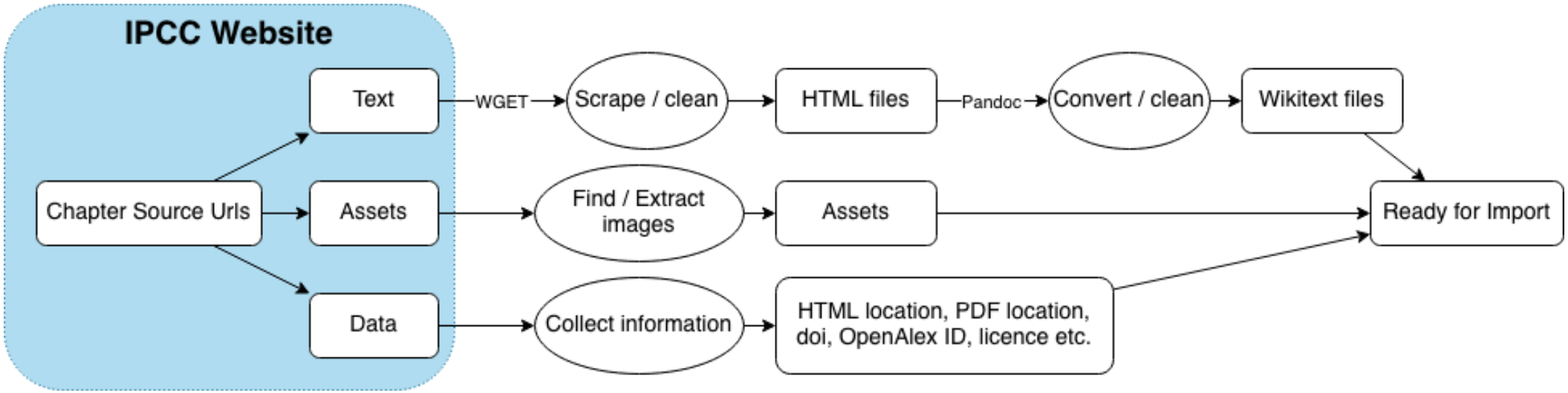
Scoping Data Sources & Data Modeling

- In-scope and out-of-scope: Only the **main text** (excluding front matter and back matter) for AR6 has currently been imported, **citations** and **data sets** have not been linked.
- Data modelling:
  - Simple (KISS) approaches of Protein Data Bank (1970s) and GeneBank (1990s) have been used to encourage community uptake.
  - Method: Bottom up / Top down / and map to schemas
  - Lit review of Wikibase KGs consulted - Zotero collection

Wikibase KG literature: Disability Wiki; EU Knowledge Graph; Enslaved.Org; Enslave Modular Ontology Modeling (MOMo); RaiseWikibase; Linking Historical Corpus Data and Annotations Using Wikibase (1737 Basque manuscript).

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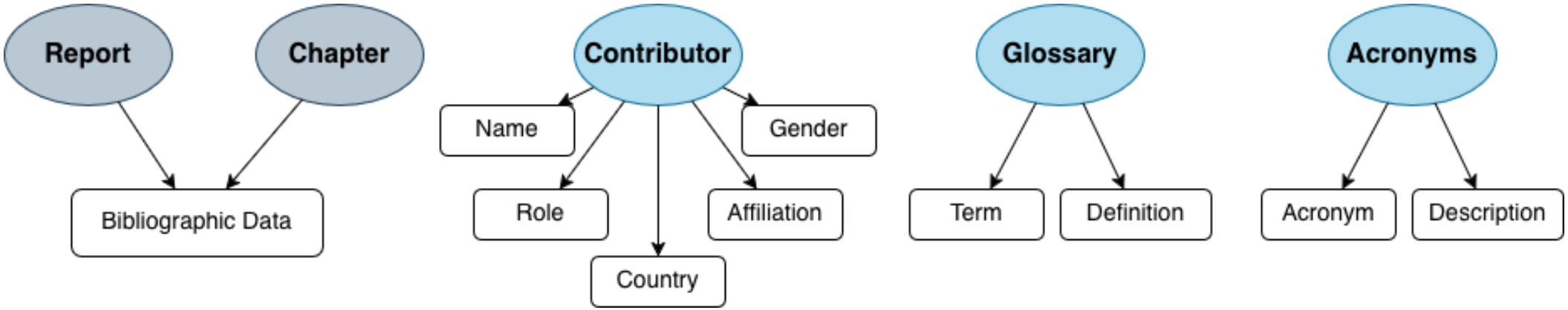
Data Retrieval



- Building a **semi-automated scraping system** from Web to MediaWiki/Wikibase
- Using the IPCC websites as a starting point, developing a method for scraping **text**, **assets**, and **additional data**
- Preparing the data for **import into a MediaWiki/Wikibase instance**

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Collecting Supporting Data



- **Additional data** from **various sources**, e.g. IPCC data - web databases, Crossref, etc.
- Preparing the supporting data for **import into Wikibase**

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Import Scrape Data — Innovation That Allows for Corpus Text and Data Import

The main innovation of the project — being able to semi-automatically web-scrape a corpus, import text and images, and make a Linked Open Data corpus backbone as an import to Wikibase and MediaWiki

Wikitext + CSV > XML > DTD > Python (WikibaseIntegrator) > Wikibase / MediaWiki

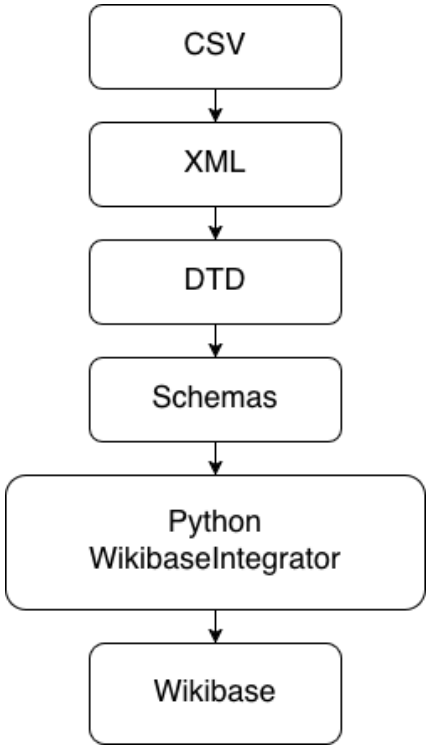
Documentation: wiki.kewl.org/projects:ckgscrape

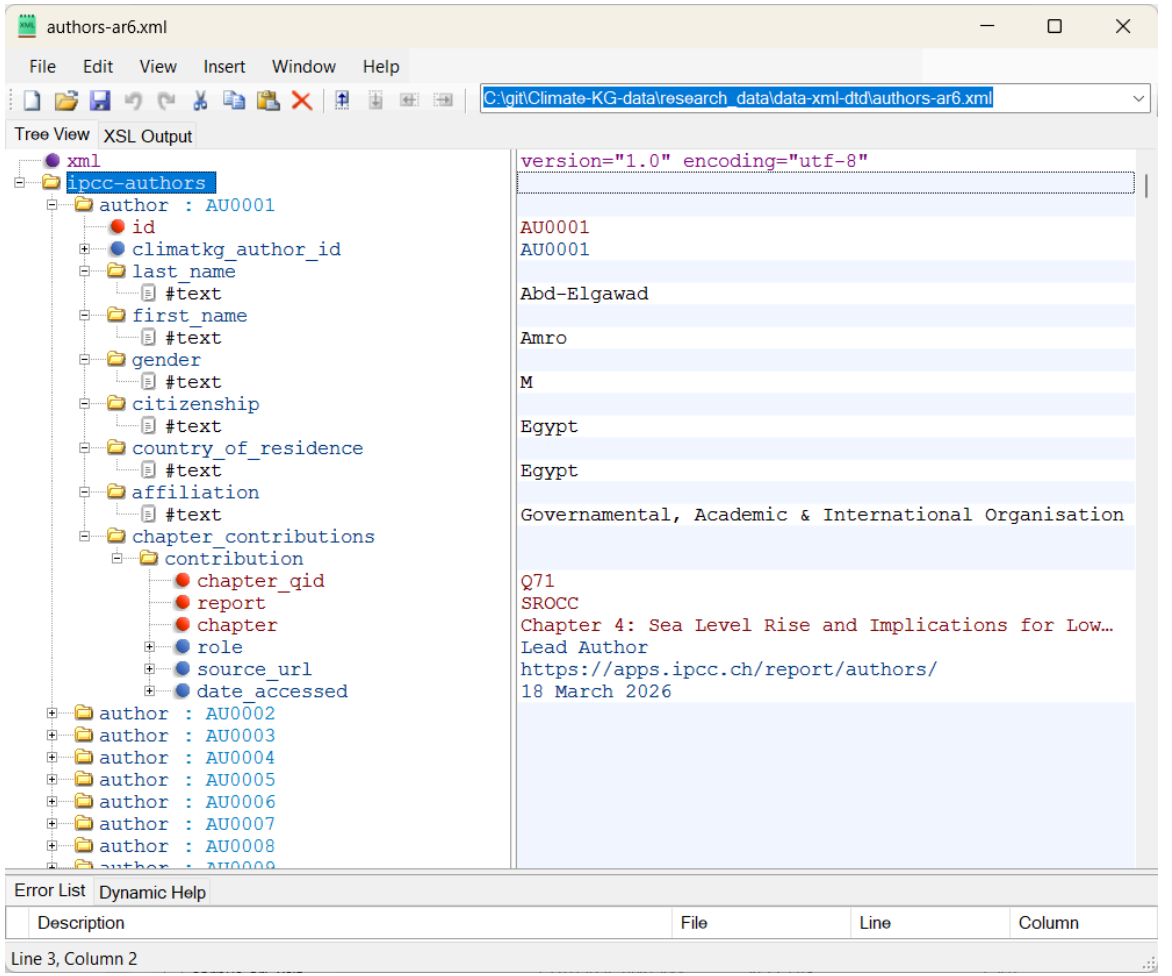
Code: Mercurial Clone `hg clone https://hg.kewl.org/pub/ckg_s2mw` (anti-bot credentials: anonymous/anonymous)

Thank you to Darron M. Broad of Runstop for the software development work.

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Import Supporting Data

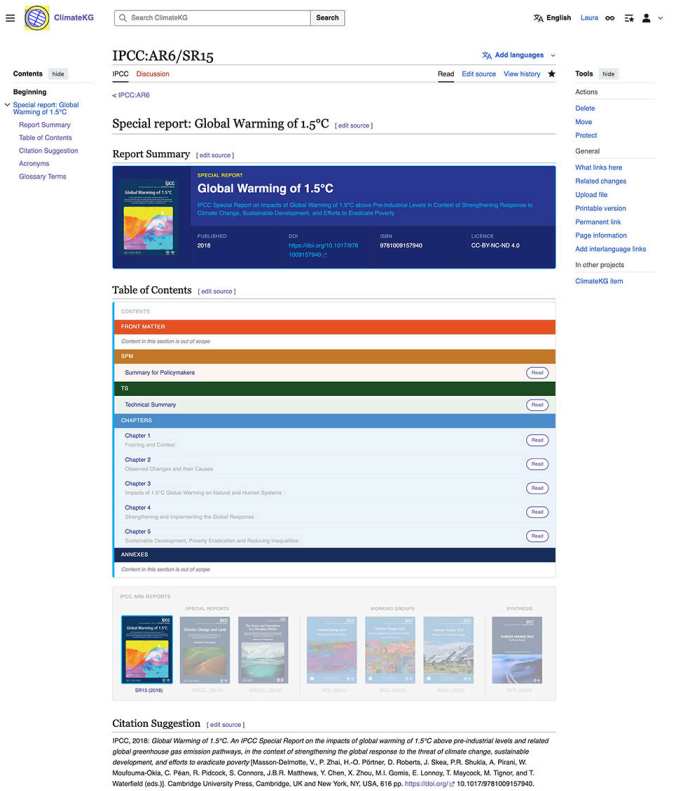


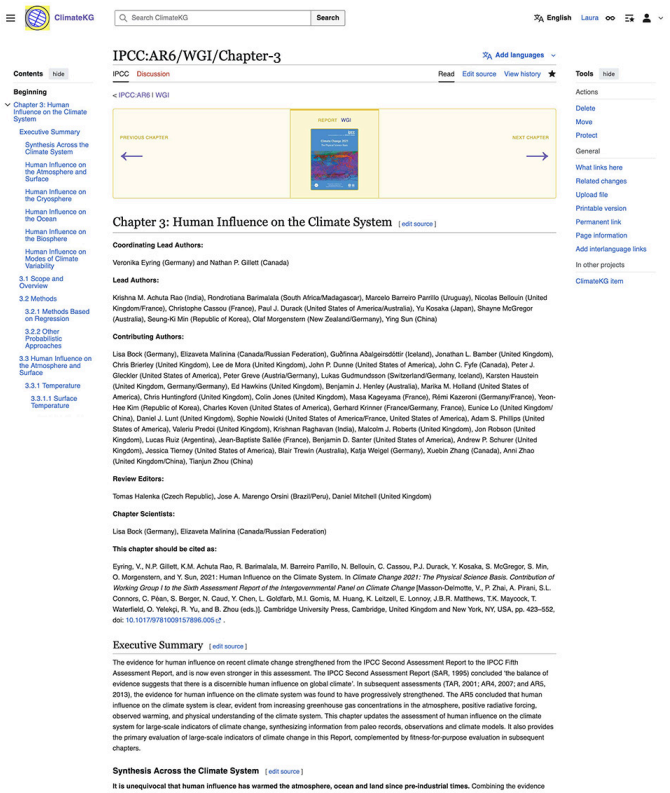


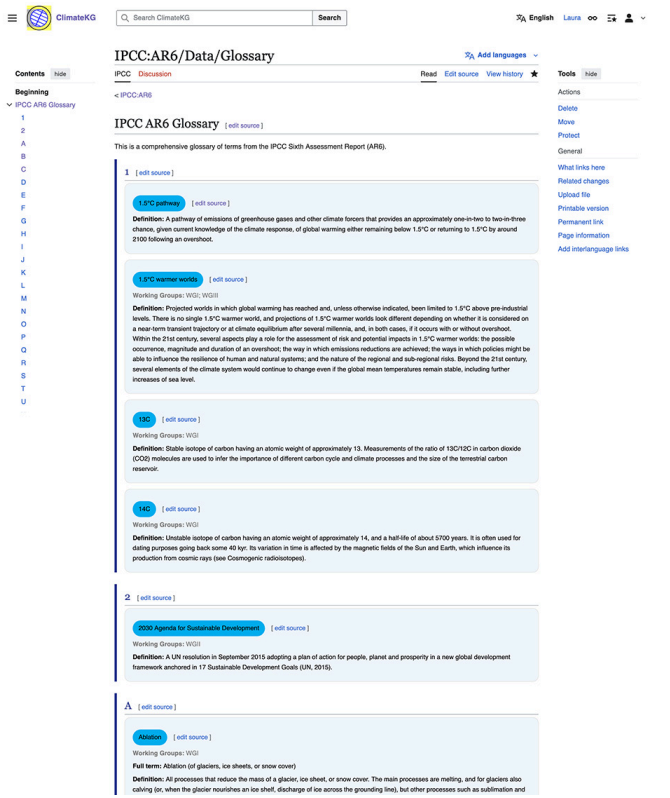
Data output as HTML using XSLT from XML: Corpus | Bibliography | Authors | Glossary | Acronyms | ClimateKG Data XML how-to

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Browsing the Corpus in MediaWiki





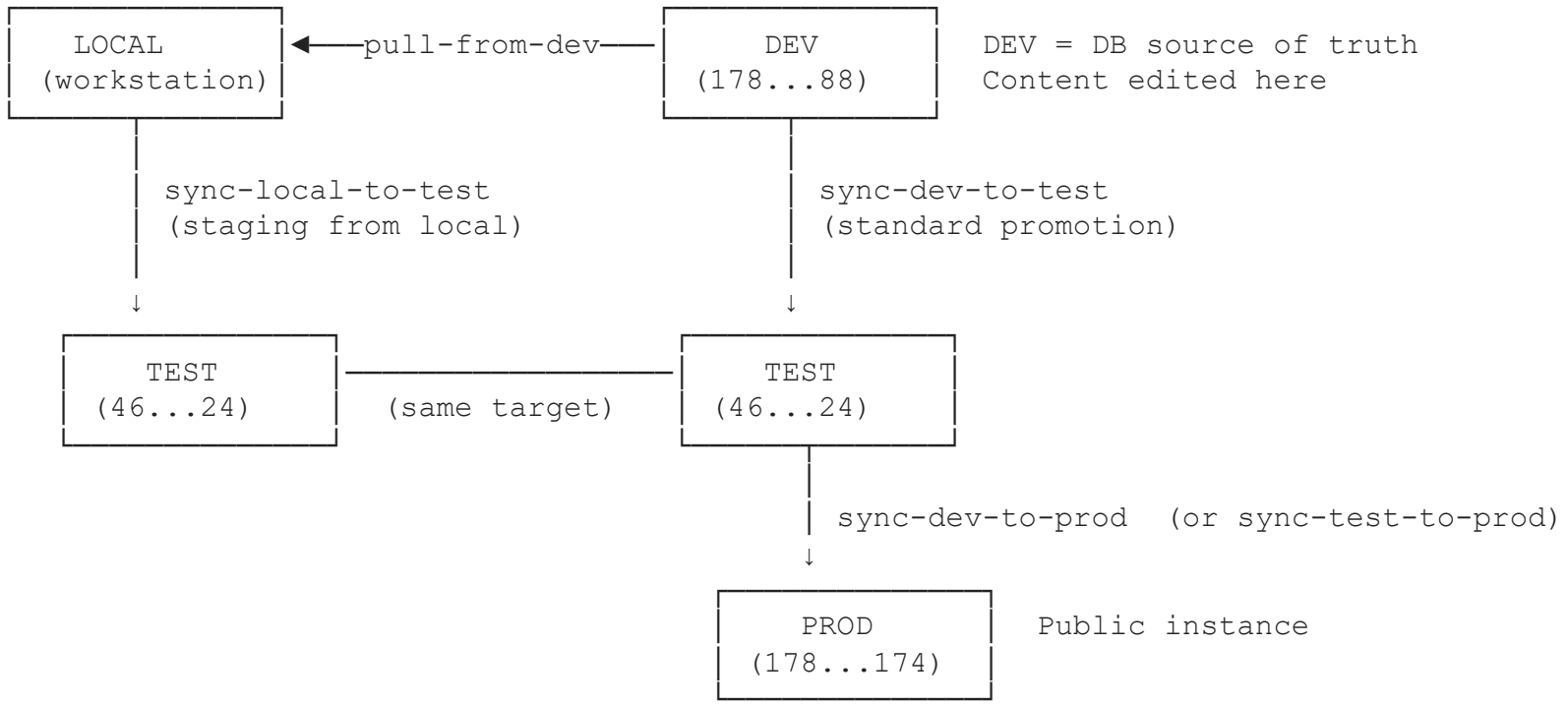


- **Cleaning** the chapter content directly in MediaWiki
- Building **navigation and browsing templates** for the corpus
- Adding **glossary** and **acronym** content to MediaWiki

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DevOps

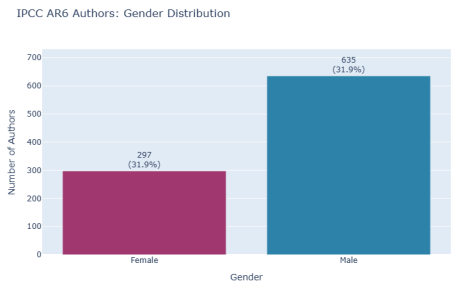
A 4 stage MediaWiki/Wikibase deployment is in place with support from #semanticClimate - this will move to WB4R and TIB hosting shortly once initial development resolved.



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Data Analysis & Visualisation

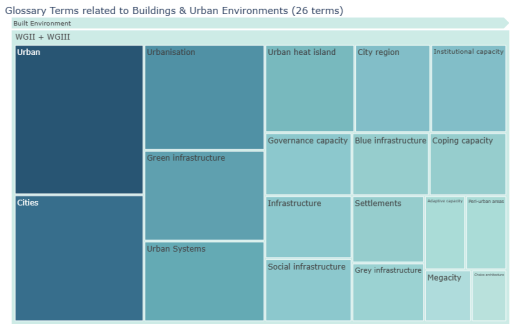
Gender Equality



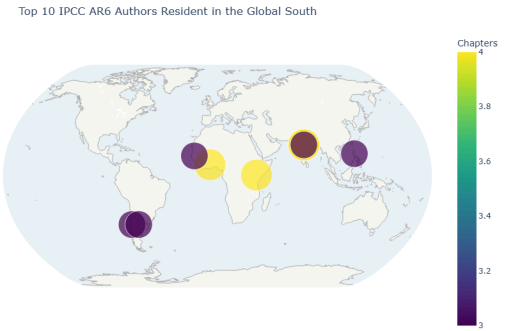
Data Dashboard



Buildings & Urban Environment



Authors Top 10 Global South



Generated direct from Wikibase KG using Python, Notebooks, and Quarto: tibhannover.github.io/ClimateKG-Data-Bench/

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CSS Paged Media as Output 📖



Chapter 9: Buildings

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9.1 Introduction

Total GHG emissions in the building sector reached 12 GtCO 2 -eq in 2019, equivalent to 21% of global GHG emissions that year, of which 57% were indirect CO 2 emissions from offsite generation of electricity and heat, followed by 24% of direct CO 2 emissions produced on-site and 18% from the production of cement and steel used for construction and/or refurbishment of buildings. If only CO 2 emissions would be considered, the share of buildings CO 2 emissions increases to 31% out of global CO 2 emissions. Energy use in residential and non-residential buildings contributed 50% and 32% respectively, while embodied emissions contributed 18% to global building CO 2 emissions. Global final energy demand from buildings reached 128.8 EJ in 2019, equivalent to 31% of global final energy demand. Residential buildings consumed 70% out of global final energy demand from buildings. Electricity demand from buildings was slightly above 43 EJ in 2019, equivalent to more than 18% of global electricity demand. Over the period 1990–2019, global CO 2 emissions from buildings increased by 50%, global final energy demand grew by 38%, with 54% increase in non-residential buildings and 32% increase in residential ones. Among energy carriers, the growth in global final energy demand was strongest for electricity, which increased by 161%.

There is growing scientific evidence about the mitigation potential of the building sector and its contribution to the decarbonisation of global and regional energy systems, and to meeting Paris Agreement goals and Sustainable Development Goals (SDGs) (IPCC, 2018 ; IEA, 2019c ; IEA 2019e). Mitigation interventions in buildings are heterogeneous in many different aspects, from building components (envelope, structure, materials, etc.) to services (shelter, heating, etc.), to building types (residential and non-residential, sometimes also called commercial and public), to building size, function, and climate zone. There are also variations between developed and developing countries in mitigation interventions to implement, as the former is challenged by the renovation of existing buildings while the latter is challenged by the need to accelerate the construction of new buildings.

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Possible **output option** Example chapter:

- Using **CSS Paged Media** to typeset the corpus content into **multi-format** output

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ClimateKG Next Steps 🚀

Planned use:

- A community resource for data science — to use and contribute
- Citizen science activities — Chapter annotation and enrichment *#ChapterChampions*
- Distribute metadata to commons (Wikidata)

Goals:

- Corpus KG service — Computation Publishing Service (CPS)
- Work with IPCC and climate community
- Become a KG repository for *all* climate literature

Thank you! Simon Worthington, Laura Oldenbourg, and team #semanticClimate — June ’26

Main </> repo: github.com/TIBHannover/climate-knowledge-graph

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